MODULE 4: SPECTROSCOPIC AND DIFFRACTION TECHNIQUES (CO: 4)

Complete Comprehensive Textbook: UV-Visible Spectrophotometry, Beer-Lambert Law, Powder XRD, Debye-Scherrer & Solved VIT FAT Question Bank

Aligned with Official VIT BACHY105 Applied Chemistry Course Syllabus, Department Question Bank & VIT FAT Papers

Syllabus Scope & Competency Checklist (Module 4):

- **Foundations of Spectroscopy:** Electromagnetic radiation interaction with matter, energy quantization ($\Delta E = h\nu = \frac{hc}{\lambda}$); molecular transitions (electronic UV-Vis, vibrational IR, rotational microwave); absorption vs emission spectra.
- **UV-Visible Spectrophotometry:** Electronic transitions ($\sigma \rightarrow \sigma^*, n \rightarrow \sigma^*, \pi \rightarrow \pi^*, n \rightarrow \pi^*$); Chromophores and Auxochromes; spectral shift nomenclature (Bathochromic / Red shift, Hypsochromic / Blue shift, Hyperchromic, Hypochromic effects).
- **Instrumentation of UV-Vis Spectrophotometer:** Dual-beam optical layout; radiation sources (Deuterium discharge lamp $190 - 350$ nm, Tungsten-halogen lamp $350 - 800$ nm); diffraction grating monochromators; quartz vs optical glass sample cuvettes; Photomultiplier Tube (PMT) detectors.
- **Beer-Lambert Law:** Rigorous mathematical derivation of Absorbance ($A = -\log_{10}\left(\frac{I_t}{I_0}\right) = \varepsilon C l$); Molar Absorptivity ($\varepsilon$); percentage transmittance ($\%T = 100 \times 10^{-A}$); real chemical and instrumental deviations from Beer's law.
- **Foundations of X-Ray Diffraction:** Dual wave-particle nature of X-rays; generation via electron bombardment on copper target (Cu K_α radiation, $\lambda = 1.5418$ \u00c5); crystalline lattices, unit cells, Miller indices (hkl), interplanar spacing (d_{hkl}).
- **Bragg's Law & Powder XRD Technique:** Derivation of path difference ($2d \sin\theta = n\lambda$); powder polycrystalline diffraction method; diffractometer geometry, 2θ scanning, peak intensity and angle interpretation.
- **Debye-Scherrer Equation & Crystallite Size:** Derivation of peak broadening ($D = \frac{K\lambda}{\beta \cos\theta}$); shape factor ($K \approx 0.9$); Full Width at Half Maximum (FWHM, β) strictly converted from degrees to radians ($\beta_{\text{rad}} = \beta_{\text{deg}} \times \frac{\pi}{180}$); instrumental vs sample broadening; crystallite size vs grain size.

1. Fundamental Concepts of Molecular Spectroscopy

1.1 The Electromagnetic Spectrum & Energy Quantization

Spectroscopy is the study of the interaction between electromagnetic radiation and matter. Electromagnetic radiation consists of mutually perpendicular oscillating electric (E) and magnetic (B) fields propagating through space at the speed of light ($c = 2.998 \times 10^8$ m/s). By Planck's quantum hypothesis, energy is absorbed or emitted only in discrete packets called **photons**:

$$\Delta E = E_2 - E_1 = h\nu = \frac{hc}{\lambda} = hc \text{ tilde}(\nu)$$

where $h = 6.626 \times 10^{-34}$ J $\cdot$ s is Planck's constant, ν is frequency (s^{-1} or Hz), λ is wavelength, and $\text{tilde}(\nu) = \frac{1}{\lambda}$ is the wavenumber (cm^{-1}).

1.2 Hierarchy of Molecular Energy Transitions

A molecule possesses translational, rotational, vibrational, and electronic energy states. By the Born-Oppenheimer approximation:

$$E_{\text{total}} = E_{\text{elec}} + E_{\text{vib}} + E_{\text{rot}} + E_{\text{trans}}$$

The energy spacing between states follows: $\Delta E_{\text{elec}} \gg \Delta E_{\text{vib}} \gg \Delta E_{\text{rot}} \gg \Delta E_{\text{trans}}$:

1. **Microwave Region ($1 - 100$ cm^{-1}):** Low energy photons ($10^{-4} - 10^{-2}$ eV) induce rotational transitions in molecules possessing a permanent dipole moment (HCl, CO).
2. **Infrared (IR) Region ($400 - 4000$ cm^{-1}):** Medium energy photons ($0.05 - 0.5$ eV) induce molecular bond vibrational stretching and bending transitions accompanied by a net change in dipole moment.
3. **Ultraviolet-Visible (UV-Vis) Region ($190 - 800$ nm or $1.5 - 6.5$ eV):** High energy photons promote outer valence bonding or non-bonding electrons into higher unoccupied antibonding molecular orbitals.

2. UV-Visible Spectrophotometry: Electronic Transitions & Shifts

2.1 Types of Electronic Transitions

When an organic molecule absorbs UV or visible light, valence electrons undergo transitions between molecular orbitals:

- $\sigma \rightarrow \sigma^*$ Transitions: High energy requirement ($\lambda < 180$ nm, Vacuum UV). Saturated alkanes (e.g., methane, ethane with $C-C, C-H$ single bonds).
- $n \rightarrow \sigma^*$ Transitions: Non-bonding lone pair electrons on heteroatoms (O, N, S, halogens) promoted to σ^* antibonding orbitals ($\lambda \approx 180-250$ nm) (e.g., alcohols, amines, alkyl halides).
- $\pi \rightarrow \pi^*$ Transitions: Absorption by unsaturated systems containing multiple bonds ($C=C, C=C-C=C$, aromatic rings). Highly probable, intense absorption bands ($\varepsilon > 10000$ L/(mol cm)) occurring at $\lambda \approx 170-220$ nm.
- $n \rightarrow \pi^*$ Transitions: Promotion of a heteroatom lone pair into an unoccupied π^* orbital (e.g., carbonyl compounds $>C=O$, nitro compounds $-NO_2$). Symmetry-forbidden by selection rules, yielding weak absorption bands ($\varepsilon \approx 10-100$ L/(mol cm)) at longer wavelengths ($\lambda \approx 280-320$ nm).

2.2 Chromophores, Auxochromes & Spectral Shifts

1. **Chromophore:** An isolated covalently unsaturated functional group capable of absorbing UV or visible radiation independently (e.g., $>C=C<$, $>C=O$, $-C\equiv N$, $-N=O$, $-NO_2$, benzene ring).
2. **Auxochrome:** A saturated substituent group containing non-bonding lone pairs that does not absorb UV radiation itself, but when attached directly to a chromophore, deepens the color and shifts absorption to longer wavelengths (e.g., $-OH$, $-NH_2$, $-OCH_3$, $-Cl$, $-SH$).
3. **Spectral Shift Terminology:**
   - **Bathochromic Shift (Red Shift):** Shift of absorption maximum (λ_{max}) to a **longer wavelength** (lower energy). Caused by extending conjugated double bonds (which narrows the HOMO-LUMO gap) or attaching electron-donating auxochromes.
   - **Hypsochromic Shift (Blue Shift):** Shift of λ_{max} to a **shorter wavelength** (higher energy). Occurs upon protonation of aniline ($C_6H_5NH_2 \rightarrow C_6H_5NH_3^+$), which locks the nitrogen lone pair, eliminating resonance with the aromatic ring.
   - **Hyperchromic Effect:** An increase in molar absorption intensity (ε_{max}).
   - **Hypochromic Effect:** A decrease in molar absorption intensity (ε_{max}).

3. The Beer-Lambert Law: Derivation & Instrumentation

3.1 Mathematical Derivation of Beer-Lambert Law

Consider a beam of monochromatic radiation of initial intensity I_0 passing through an absorbing solution of concentration C contained in an optical cell of pathlength l:

- **Lambert's Law (1760):** The fraction of light absorbed is directly proportional to the thickness of the medium (dx), independent of initial intensity:

$$-\frac{dI}{I} = k_1 dx$$

- **Beer's Law (1852):** The fraction of light absorbed is directly proportional to the number of absorbing molecules per unit volume (molar concentration C):

$$-\frac{dI}{I} = k_2 C$$

Combining both laws into the differential rate of attenuation:

$$-\frac{dI}{I} = k' C dx$$

Integrating over the pathlength from $x = 0$ (where $I = I_0$) to $x = l$ (where $I = I_t$):

$$-\int_{I_0}^{I_t} \frac{dI}{I} = k'C \int_0^l dx \quad \Longrightarrow \quad -\ln\left(\frac{I_t}{I_0}\right) = \ln\left(\frac{I_0}{I_t}\right) = k'Cl$$

Converting natural logarithm to common base-10 logarithm ($\ln X = 2.303 \log_{10} X$):

$$\log_{10}\left(\frac{I_0}{I_t}\right) = \left(\frac{k'}{2.303}\right) Cl = \varepsilon Cl$$

Defining **Absorbance (A)** or Optical Density (OD):

$$A = \log_{10}\left(\frac{I_0}{I_t}\right) = \varepsilon C l$$

where:
- A: Dimensionless absorbance.
- ε: Molar Absorptivity or Molar Extinction Coefficient (L mol^{-1} cm^{-1}), a characteristic physical constant of the absorbing analyte at wavelength λ.
- C: Molar concentration of analyte (mol/L).
- l: Optical pathlength of the sample cell (cm, standard $l = 1.00$ cm).
- **Transmittance (T):** Fraction of incident light transmitted through the solution:

$$T = \frac{I_t}{I_0} = 10^{-A} \quad \Longrightarrow \quad \%T = 100 \times T = 100 \times 10^{-A}$$

$$A = -\log_{10} T = -\log_{10}\left(\%\frac{T}{100}\right) = 2 - \log_{10}(\%T)$$

3.2 Instrumentation of Dual-Beam UV-Vis Spectrophotometer

A modern double-beam spectrophotometer splits the monochromatic light into two identical beams, simultaneously passing one through the sample cuvette and the other through an identical reference/solvent blank cuvette:

1. **Light Sources:**
   - **Deuterium (D_2) Discharge Lamp:** Provides intense continuous emission in the UV region ($190 - 350$ nm).
   - **Tungsten-Halogen Filament Lamp:** Provides continuous radiation across the Visible and Near-IR range ($350 - 1100$ nm).
2. **Monochromator:**
   - Consists of an entrance slit, collimating mirror, reflection diffraction grating (typically $1200 - 2400$ grooves/mm), and focusing exit slit to isolate narrow monochromatic wavelength bands ($\Delta\lambda < 1$ nm).
3. **Optical Cuvettes (Sample Cells):**
   - **Quartz (Fused Silica) Cuvettes:** Mandatory for UV measurements below 350 nm because optical glass and plastics absorb strongly in the UV spectrum.
   - **Optical Glass Cuvettes:** Suitable strictly for visible measurements ($350 - 800$ nm).
4. **Detectors:**
   - **Photomultiplier Tube (PMT):** Photoelectric cathode emits primary photoelectrons, which strike a series of secondary dynodes, cascading an amplification factor of $10^6 - 10^7$ electrons per photon, offering extreme sensitivity for trace detection.

4. Powder X-Ray Diffraction (PXRD) & Debye-Scherrer Crystallite Sizing

4.1 X-Ray Production & Bragg's Law

X-rays are high-energy electromagnetic radiation with wavelengths of the same order of magnitude as interatomic spacings in crystalline solids ($\lambda \approx 0.5 - 2.5$ \u00c5 or $0.05 - 0.25$ nm).

- **Bragg's Law Derivation (William Lawrence Bragg, 1913):** Consider a collimated monochromatic X-ray beam incident at grazing angle θ on parallel crystal lattice planes with Miller indices (hkl) separated by interplanar distance d: Ray 1 reflects from the first atomic plane at angle θ. Ray 2 penetrates and reflects from the second atomic plane at depth d. The geometric extra pathlength traveled by Ray 2 relative to Ray 1 is $AB + BC = d\sin\theta + d\sin\theta = 2d\sin\theta$. For constructive interference and intense diffraction peaks to emerge, this path difference must equal an integer number of complete wavelengths ($n\lambda$):

$$2d\sin\theta = n\lambda$$

  where n is the diffraction order ($n = 1, 2, ...$), d is interplanar spacing, θ is Bragg angle, and λ is wavelength (typically 1.5418 \u00c5 for Cu K_α).
- **Interplanar Spacing for Cubic Crystals:**

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

where a is the cubic unit cell edge length.

4.2 The Powder Method & Diffractometer Recording

In Powder XRD, the sample is ground into an ultrafine powder containing millions of randomly oriented crystalline grains. Statistically, for every family of (hkl) planes, thousands of crystallites are oriented at the exact Bragg angle θ to the incident beam, forming concentric **Debye-Scherrer diffraction cones**. The detector rotates around the sample through an angle of 2θ, recording X-ray counts per second as a function of scattering angle 2θ. The resulting diffractogram exhibits sharp peaks whose positions (2θ) identify the crystal phase and lattice parameters, while peak intensities indicate atomic positions within the unit cell.

4.3 The Debye-Scherrer Equation for Crystallite Size

In an ideal infinite crystal, diffraction peaks are infinitely sharp mathematical delta functions. In real nanostructured materials where crystallite size $D < 100$ nm, the finite number of atomic planes results in incomplete destructive interference at angles slightly away from θ, causing physical **broadening of diffraction peaks**.

- **Paul Scherrer's Formula (1918):**

$$D = \frac{K\lambda}{\beta \cos\theta}$$

where:

- D: Mean crystallite size (average diameter of single-crystalline coherent scattering domains) along the direction perpendicular to the reflecting plane.
- K: Dimensionless shape factor (typically $K = 0.90$ for spherical crystallites with cubic symmetry, $K = 0.94$ for FWHM).
- λ: X-ray wavelength (0.15418 nm for Cu K_α).
- θ: Bragg diffraction angle (one-half of the recorded detector peak position 2θ).
- β: The **Full Width at Half Maximum (FWHM)** of the diffraction peak measured in **radians**.
- **Critical Calculation Rule:** Diffractometer software reports FWHM in **degrees (2θ)**. It must be converted into radians before substitution into the Scherrer equation:

$$\beta_{\text{rad}} = \beta_{\text{deg}} \times \frac{\pi}{180} = \beta_{\text{deg}} \times \left(\frac{3.14159}{180}\right)$$

- **Instrumental Broadening Correction:** The observed experimental peak width β_{obs} contains instrumental broadening β_{inst} caused by finite X-ray source slit width and beam divergence. True sample broadening is obtained via Gaussian deconvolution:

$$\beta_{\text{true}}^2 = \beta_{\text{obs}}^2 - \beta_{\text{inst}}^2$$

VIT FAT Solved Question Bank: Step-by-Step Numerical Derivations

FAT Problem 1: Debye-Scherrer Crystallite Size Determination from Powder XRD

Question (10 Marks): A synthesized titanium dioxide (TiO_2, anatase) nanopowder is characterized by Powder X-Ray Diffraction using Cu K_α radiation ($\lambda = 1.5418$ \u00c5 $= 0.15418$ nm). The prominent (101) anatase diffraction peak is recorded at a detector angle of $2\theta = 25.32°$. The Full Width at Half Maximum (FWHM) of this peak is measured to be $\beta_{obs} = 0.420°$. The instrumental broadening determined using a standard silicon reference is $\beta_{inst} = 0.080°$. Taking the Scherrer shape factor as $K = 0.90$: ("i") Calculate the Bragg diffraction angle θ in degrees and radians. ("ii") Correct for instrumental broadening to find the true physical sample broadening β_{true} in degrees and convert it strictly into radians. ("iii") Using the Debye-Scherrer equation, compute the average crystallite size D of the TiO_2 nanoparticles in nanometers. ("iv") Calculate the interplanar spacing d_{101} in Angstroms using Bragg's law for first-order diffraction ($n = 1$).

Complete Step-by-Step Solution:

Step 1: Calculate Bragg Angle θ:

$$2\theta = 25.32° \implies \theta = \frac{25.32}{2} = 12.66°$$

$$\cos\theta = \cos(12.66°) = 0.97568$$

Step 2: Correct Peak Broadening & Convert to Radians: Applying Gaussian deconvolution correction:

$$\beta_{true} = \sqrt{\beta_{obs}^2 - \beta_{inst}^2} = \sqrt{(0.420)^2 - (0.080)^2} = \sqrt{0.1764 - 0.0064} = \sqrt{0.1700} = 0.4123°$$

Convert degrees to radians:

$$\beta_{rad} = \beta_{true} \times \left(\frac{\pi}{180}\right) = 0.4123 \times \left(\frac{3.14159}{180}\right) = 0.4123 \times 0.017453 = 7.196 \times 10^{-3} \text{ radians}$$

Step 3: Debye-Scherrer Crystallite Size Calculation:

$$D = \frac{K\lambda}{\beta_{rad}\cos\theta}$$

$$D = \frac{0.90 \times 0.15418 \text{ nm}}{(7.196 \times 10^{-3}) \times 0.97568} = \frac{0.13876 \text{ nm}}{7.021 \times 10^{-3}} = 19.764 \text{ nm}$$

Step 4: Interplanar Spacing d_{101} using Bragg's Law:

$$2d\sin\theta = n\lambda \implies d = \frac{\lambda}{2\sin\theta}$$

$$\sin\theta = \sin(12.66°) = 0.21919$$

$$d_{101} = \frac{1.5418 \text{ \u00c5}}{2 \times 0.21919} = \frac{1.5418}{0.43838} = 3.517 \text{ \u00c5} = 0.3517 \text{ nm}$$

(Matches standard ICDD reference for anatase (101) planes at $d = 3.52$ \u00c5).

Boxed Final Answers:

$$\theta = 12.66°, \quad \beta_{rad} = 7.20 \times 10^{-3} \text{ rad}, \quad D = 19.76 \text{ nm}, \quad d_{101} = 3.52 \text{ \u00c5}.$$

FAT Problem 2: Beer-Lambert Law Quantitative Photometric Analysis

Question (10 Marks): A standard solution of potassium permanganate ($KMnO_4$, molar mass 158.03 g/mol) of concentration $C_1 = 4.0 \times 10^{-5}$ M exhibits a transmittance of $T_1 = 45.0\%$ at its absorption maximum $\lambda_{max} = 525$ nm in an optical cuvette of pathlength $l = 1.00$ cm. ("i") Calculate the absorbance A_1 of this standard solution. ("ii") Determine the molar absorptivity ε of $KMnO_4$ at 525 nm with appropriate units. ("iii") An industrial wastewater effluent sample containing an unknown concentration of permanganate is analyzed under identical conditions and exhibits an absorbance of $A_2 = 0.625$. Calculate the molar concentration of permanganate in the wastewater. ("iv") Express the permanganate concentration in the wastewater in milligrams per liter (mg/L" or ppm").

Complete Step-by-Step Solution:

Step 1: Calculate Standard Absorbance A_1:

$$A_1 = 2 - \log_{10}(\%T) = 2 - \log_{10}(45.0) = 2 - 1.6532 = 0.3468$$

Step 2: Determine Molar Absorptivity ε: From Beer-Lambert law ($A = \varepsilon C l$):

$$\varepsilon = \frac{A_1}{C_1 \times l} = \frac{0.3468}{(4.0 \times 10^{-5}\ \text{mol/L}) \times 1.00\ \text{cm}} = \frac{0.3468}{4.0 \times 10^{-5}} = 8670.0\ \text{L mol}^{-1}\ \text{cm}^{-1}$$

Step 3: Determine Unknown Wastewater Concentration:

$$C_2 = \frac{A_2}{\varepsilon \times l} = \frac{0.625}{8670.0 \times 1.00} = 7.2088 \times 10^{-5}\ \text{mol/L}$$

Step 4: Express Concentration in ppm (mg/L):

$$\text{Mass Concentration} = C_2 \times M_{KMnO_4} = (7.2088 \times 10^{-5}\ \text{mol/L}) \times (158.03\ \text{g/mol})$$

$$\text{Mass Concentration} = 0.01139\ \text{g/L} = 11.39\ \text{mg/L} = 11.39\ \text{ppm}$$

Boxed Final Answers:

$$A_1 = 0.347, \quad \varepsilon = 8670\ \text{L/(mol cm)}, \quad C_{\text{unknown}} = 7.21 \times 10^{-5}\ \text{M}, \quad \text{Concentration} = 11.39\ \text{mg/L (ppm)}.$$